

Verification of the Laws of Refraction of Light

Aim

To measure the angle of incidence, the angle of refraction, the angle of emergence and verify the laws of refraction of light.

Requirements

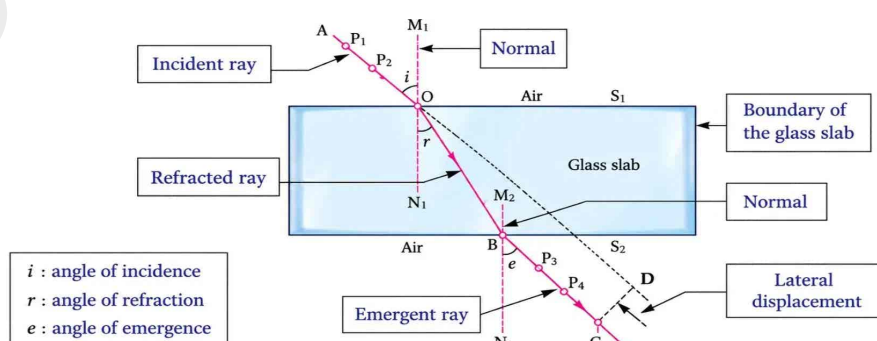
A rectangular glass slab, a drawing board, a drawing paper, drawing pins, pins, and a protractor.

Procedure

1. Fix the drawing paper onto the drawing board using drawing pins.
2. Place the rectangular glass slab on the paper and trace its boundary with a sharp pencil.
3. Remove the glass slab and draw a normal M_1N_1 at point O. Draw a line AO that makes an angle of incidence (i) of 30° with the normal M_1N_1 .
4. Fix pins P_1 and P_2 on line AO so that the pins stand vertical and are well separated.
5. Place the glass slab back in its original position. Looking through the glass slab from the opposite side S_2 , view the images of the bases of pins P_1 and P_2 , and fix two more pins P_3 and P_4 such that their bases line up exactly with these images.
6. Remove pins P_3 and P_4 and mark their positions with small circles. Draw a line BC through points P_3 and P_4 . Remove pins P_1 and P_2 along with the glass slab. Join OB and draw a normal M_2N_2 at point B.
7. Measure the angle of incidence (i), the angle of refraction (r), and the angle of emergence (e).
8. Repeat the same procedure for angles of incidence of 45° and 60° .
9. Work out the ratio $\sin i / \sin r$ for each case.

Diagram

(Label the various components in the given diagram – Refraction of light through a glass slab)



Observation

The path of the ray for angle of incidence of $30^\circ/45^\circ/60^\circ$ is AOBC.

Observation Table

Angle of Incidence (i)	Angle of Refraction (r)	Angle of Emergence (e)	$\sin i / \sin r$
30°	20°	30°	1.46
45°	30°	45°	1.41
60°	38°	60°	1.40

Inferences / Conclusions

1. Incident ray AO and emergent ray BC are parallel to each other. The lateral shift is equal to CD. (This shift depends upon the thickness t of the glass slab, the refractive index of glass and the angle of incidence.)
2. The angle of incidence (i) > the angle of refraction (r). This means when a ray of light travels from air to glass, it bends towards the normal.
3. The angle of incidence (i) = the angle of emergence (e).
4. The incident ray and the refracted ray are on the opposite side of the normal to the surface at the point of incidence, and all the three, i.e., the incident ray, the refracted ray and the normal, are in the same plane.
5. For a given pair of media, the ratio of the sine of the angle of incidence to the sine of the angle of refraction is constant (Snell's law). This constant is called the refractive index of the second medium with respect to the first medium.

Precaution

Pins should be fixed such that they are vertical and well separated.

Multiple Choice Questions

1. Rakesh performs the experiment on tracing the path of a ray of light passing through a rectangular glass slab for different angles of incidence. He observes that in all cases

- (A) $i > r$ but $i = e$ (B) $i < r$ but $i = e$
(C) $i > e$ but $i = r$ (D) $i < e$ but $i = r$

Answer: (A) $i > r$ but $i = e$

2. When a ray of light travels from air to glass and strikes the surface of separation at 90° with the surface,

- (A) it bends towards the normal (B) it bends away from the normal
(C) it passes without bending (D) it passes randomly

Answer: (C) it passes without bending

3. If a ray of light incident from a denser medium passes through a rarer medium in a straight line, the angle of incidence must be

- (A) 0° (B) 30° (C) 60° (D) 90°

Answer: (A) 0°

Remember

The lateral displacement depends on the thickness of the glass, angle of incidence and angle of refraction.

Date: _____

Teacher's Signature: _____